



Operating Systems & Networks

CTEC1704C

Lecture - 6 : Logic Gates & Hardware Memory

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Outline

- Logic Gates
- Boolean Operations: AND
- Boolean Operations: OR
- Boolean Operations: NOT
- Boolean Operations: NAND
- Boolean Operations: NOR
- Memory Technologies



Logic gates are the fundamental building blocks in digital electronics"

- Used to perform logical operations based on the inputs provided to it and gives a logical output that can be either high (1) or low (0)"
- The operation of logic gates is based on Boolean algebra or mathematics"
- There are basically seven main types of logic gates that are used to perform various logical operations in digital systems"
- By combining different logic gates, complex operations are performed and circuits like flip-flops, counters and processors are designed" /n this article (we will see various types of logic gates in detail"
- Logic gates find their uses in our day-to-day lives such as in the architecture of our telephones, laptops, tablets and memory devices"



Transistors

- A transistor is an on/off switch
 - On allows current to flow through the circuit
 - Off blocks the flow of current
- We can create a **physical** equivalent of the **logical** operations AND/OR/NOT by combining a few transistors





Boolean Operations: AND

- Yields true binary value & '1' if and only if both of its operands are true
- Represented by simple concatenation or the dot operator:
 $A \cdot B$ or AB
- Symbol in a circuit diagram: 
- Truth table:

Inputs		Output
A	B	F = A AND B
0	0	0
0	1	0
1	0	0
1	1	1

Source: <https://en.wikipedia.org/wiki/File:NMO69AND9gate.png>



Boolean Operations: OR

- Yields true if either or both of its operands are true
- Represented by the plus sign: $A + B$

- Symbol in a circuit diagram: 

- Truth table:

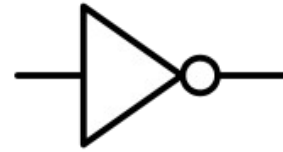
Inputs		Output
A	B	$F = A \text{ OR } B$
0	0	0
0	1	1
1	0	1
1	1	1



Boolean Operations: NOT

- Inverts the value of its operand
- Represented by the operator: $\bar{A}$
- Symbol in a circuit diagram:
- Truth table:

$\bar{A}$



Input	Output
A	F = NOT A
0	1
1	0

- We can combine the basic Boolean operations into more complex functions
- For example: NAND
- Symbol in a circuit diagram:
- Truth table:



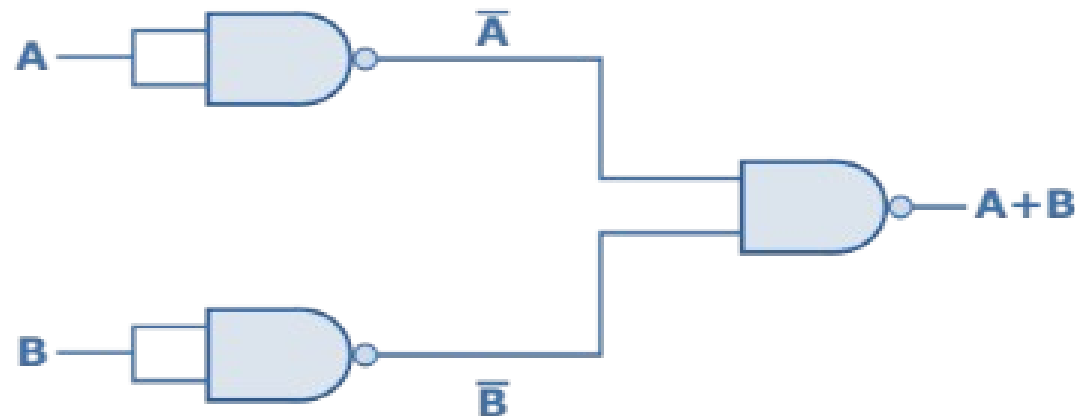
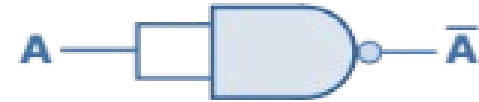
Inputs		Output
A	B	$F = A \text{ NAND } B$
0	0	1
0	1	1
1	0	1
1	1	0

Source: https://en.wikipedia.org/wiki/File:NAND_gate.svg



Implementing NAND Gates

- AND* OR* and NOT can be implemented using only NAND Gates
- Great for circuit hard (are: only need one type of gate)
- NAND gates are **functionally complete**



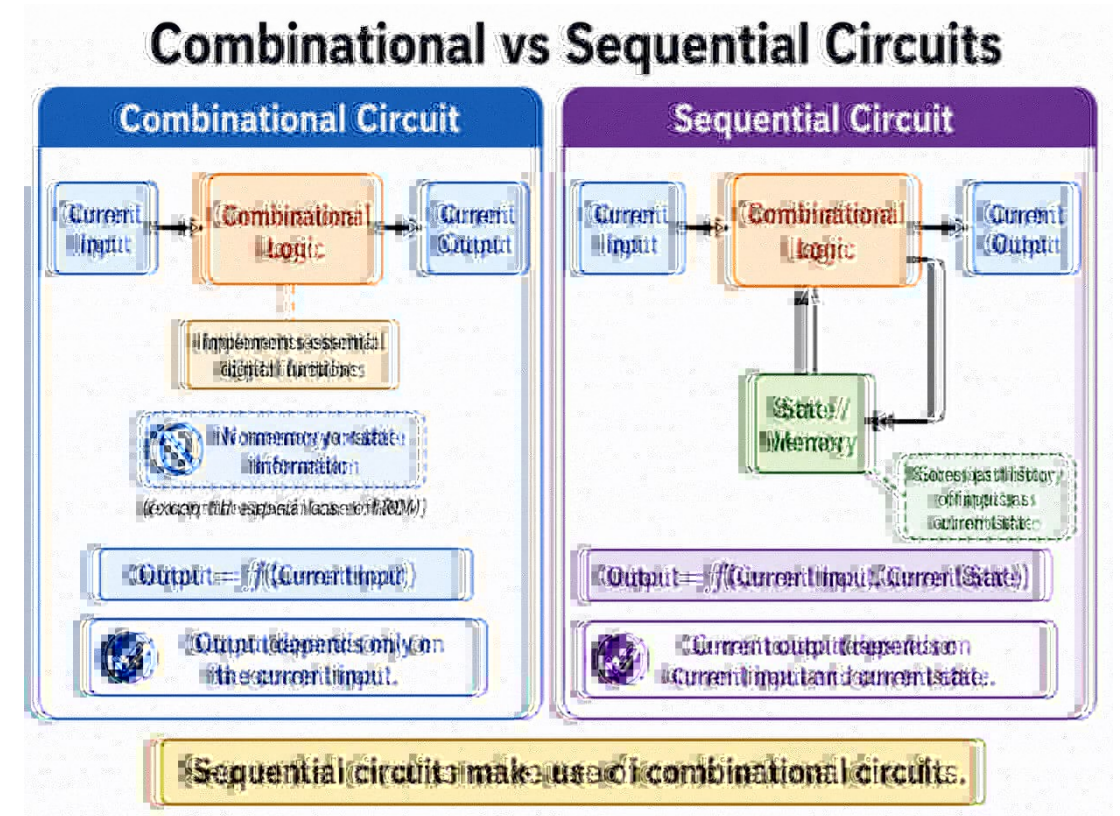


NOR gate



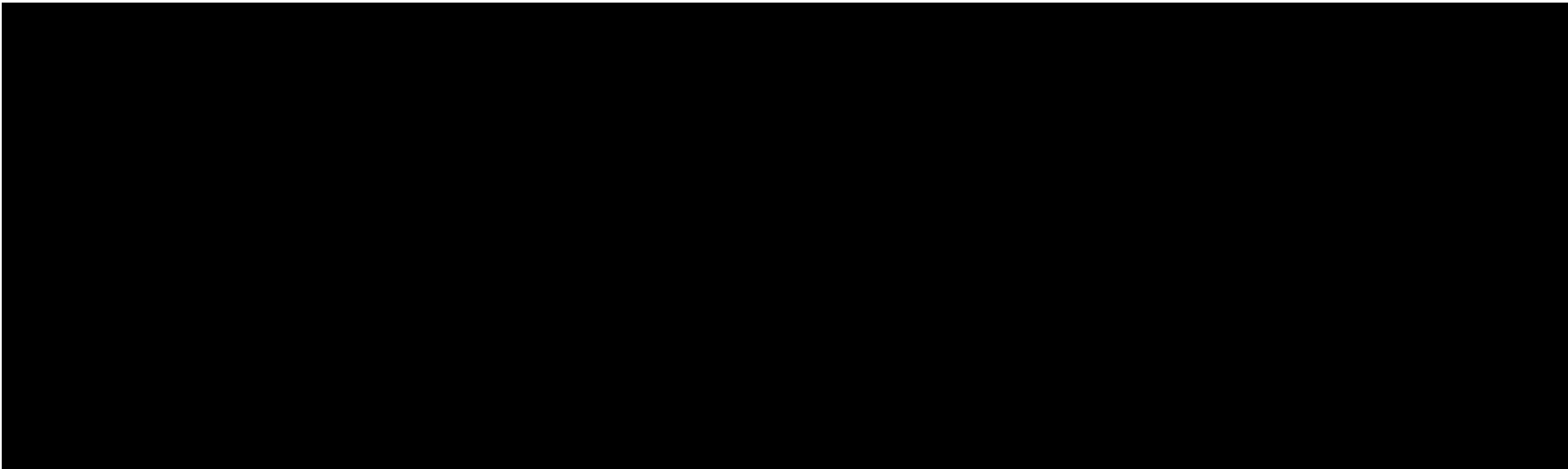
A	B	Out
0	0	1
0	1	0
1	0	0
1	1	0

- ;om inat onal circuits depend only on current input
 - cannot store state informat on<
- 6e3uent al circuits:
 - ;urrent output depends not only on the current input* ut also on the past history of inputs
 - #ses com inat onal circuits as uilding loc!s
 - #seful se3uent al circuits:
 - 8lip.-ops





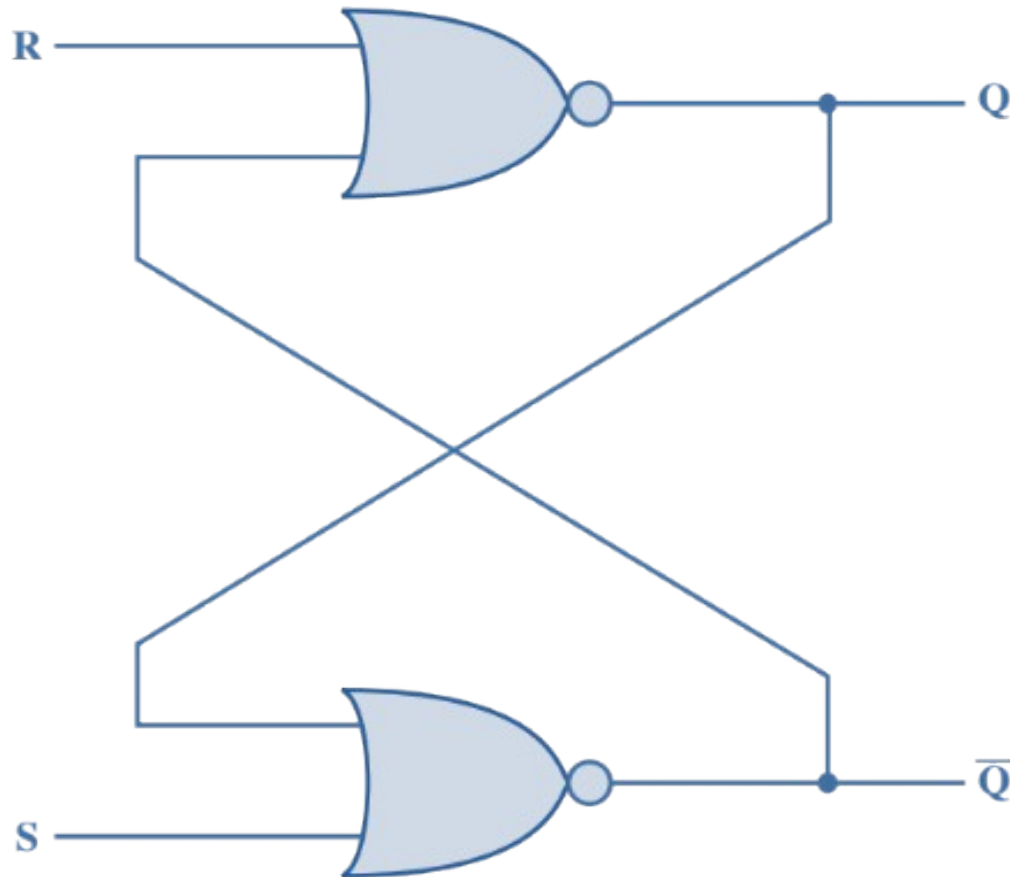
- Simplest form of sequential circuit
- There are a variety of flip-flops all of which share the following properties:
 - The flip-flop is a **bistable** device" /t e, ists in **one of two states** and* in the absence of input* remains in that state" Thus* the flip-flop can function as a **1-bit memory**"
 - The flip-flop has two outputs* which are always the complements of each other"





=, ample 8lip.8lop: the 6.R Latch

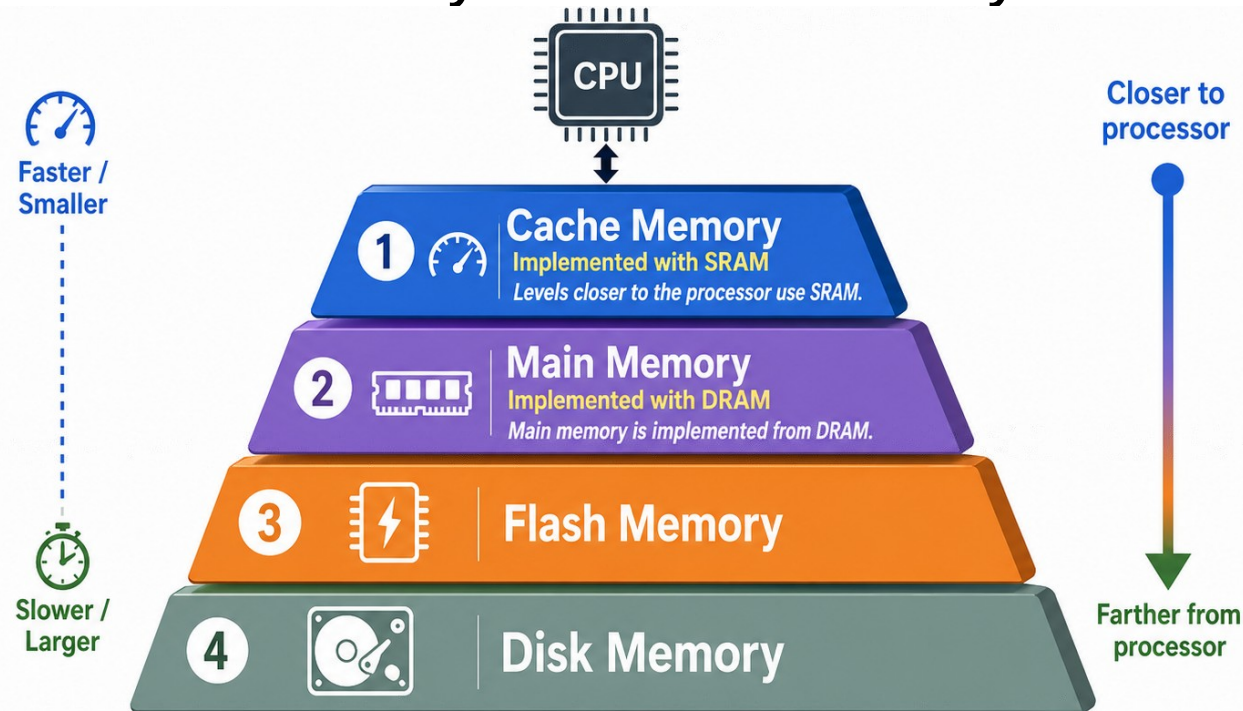
- >air of cross.coupled NOR gates
- /nputs 6 %set' and R %reset'



S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	-

Not allo (ed<

- There are four primary technologies used today in memory hierarchies
- Main memory is implemented from DRAM %dynamic random access memory'* (hile le\$els closer to the processor %caches' use 6RAM %sta t c random access memory'
- The Third and 8ourth are 8lash memory and dis! memory"



- 6RAMs are simply integrated circuits that are memory arrays (with 'usually' a single access port that can provide either a read or a write)
- 6RAMs have a 0,ed access time to any datum* though the read and write access times may differ
- 6RAMs typically use six to eight transistors per bit to present the information from being disturbed (when read)



- In a SRAM, as long as power is applied, the value can be kept indefinitely. In a dynamic RAM (DRAM), the value kept in a cell is stored as a charge in a capacitor.
- A single transistor is then used to access this stored charge, either to read the value or to overwrite the charge stored there.
- As DRAMs store the charge on a capacitor, it cannot be kept indefinitely and must periodically be refreshed. That is why this memory structure is called dynamic, in contrast to the static storage in an SRAM cell.





Differences

<u>SRAM</u>	<u>DRAM</u>
1. SRAM has lower access time, so it is faster compared to DRAM.	1. DRAM has higher access time, so it is slower than SRAM.
2. SRAM is costlier than DRAM.	2. DRAM costs less compared to SRAM.
3. SRAM requires constant power supply, which means this type of memory consumes more power.	3. DRAM offers reduced power consumption, due to the fact that the information is stored in the capacitor.
4. Due to complex internal circuitry, less storage capacity is available compared to the same physical size of DRAM memory chip.	4. Due to the small internal circuitry in the one-bit memory cell of DRAM, the large storage capacity is available.
5. SRAM has low packaging density.	5. DRAM has high packaging density.



8lash and dis! memory

- Flash memory is a type of electrically erasable programmable read-only memory (EEPROM).
- It is a type of nonvolatile memory that erases data in units called blocks and rewrites data at the byte level
- More on Flash Memory –
- <https://computer.howstuffworks.com/flash-memory.htm>
- What you should know about disk storage
- <https://computer.howstuffworks.com/hard-disk.htm>,
<https://www.youtube.com/watch?v=NtPc0jI21i0>
(interesting)



